Two-Sample Test for Laws of Random Probabilities via Optimal Transport

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Joint work with

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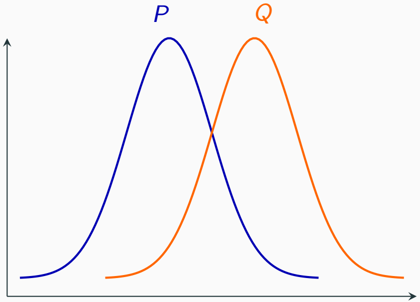
LUISS
Università di Roma

Hugo Lavenant

Two-Sample Testing in $\mathbb{R}$



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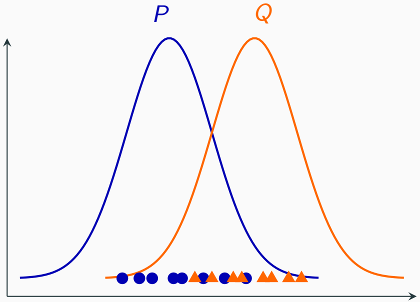

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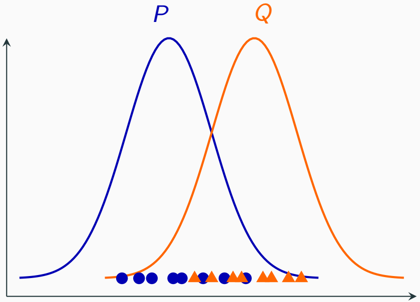

Do they come from the same law?

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(unknown) $P, Q \in \mathcal{P}(\mathbb{R})$

(Observed) $X_1, \ldots, X_n \overset{\text{i.i.d}}{\sim} P, \quad Y_1, \ldots, Y_n \overset{\text{i.i.d}}{\sim} Q.$

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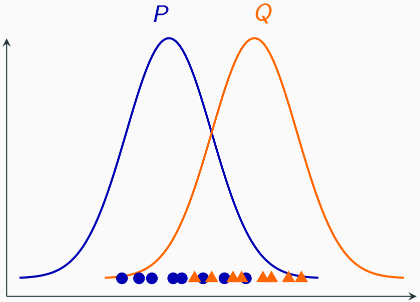

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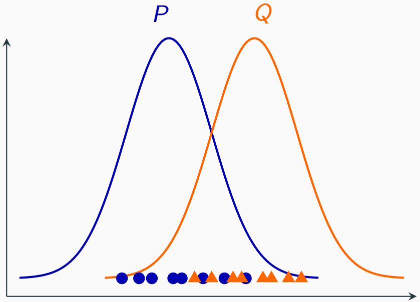

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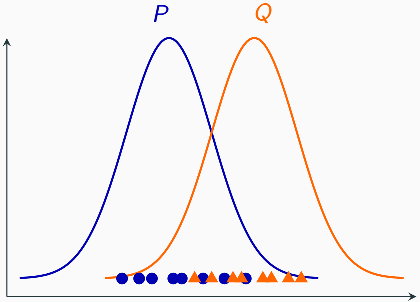

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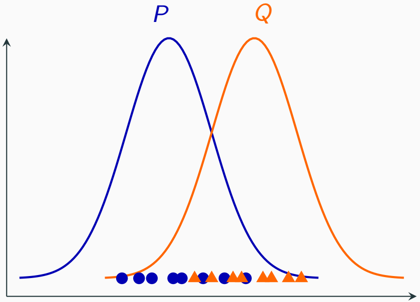

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$$d\left(\frac{1}{n} \sum_{i=1}^{n} \delta_{X_i}, \frac{1}{n} \sum_{i=1}^{n} \delta_{Y_i} \right) \text{ is large}$$

Two-Sample Testing in $\mathbb{R}^d$

Works in $\mathbb{R}^d$:

- Gretton et al. (2012) — maximum mean discrepancy
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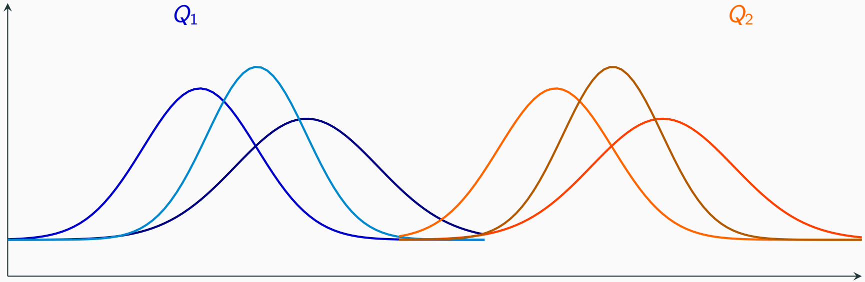
What about **beyond** $\mathbb{R}^d$? What if the observed samples are themselves **probability measures**?

Two-sample testing in $\mathcal{P}(\mathcal{P}(\mathbb{X}))$



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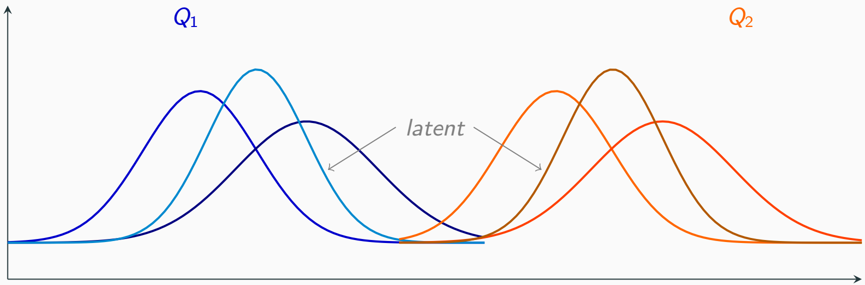

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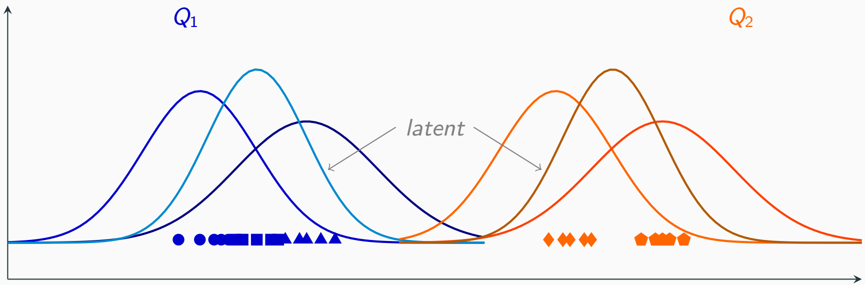

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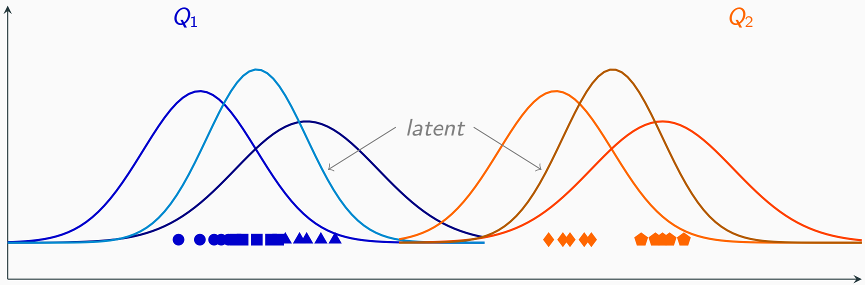

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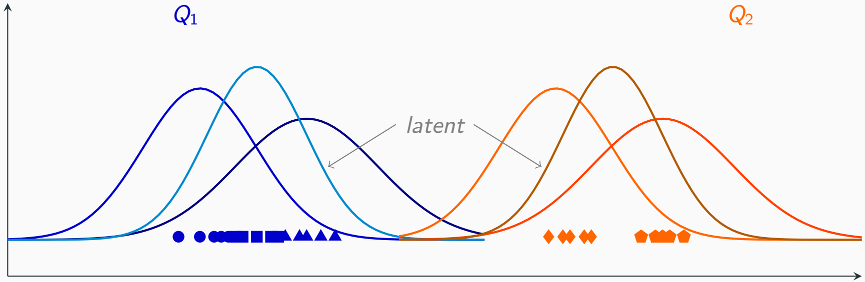

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$\rightarrow$ Need to work at the level of **Random Probability Measures**

Hierarchical sample

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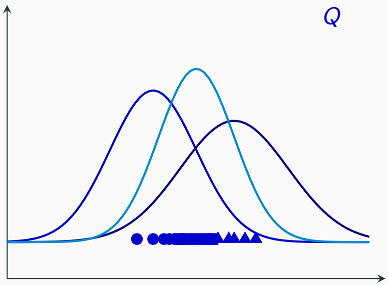
(latent) $\widetilde{P}_1, \ldots, \widetilde{P}_n \overset{\text{i.i.d}}{\sim} Q$



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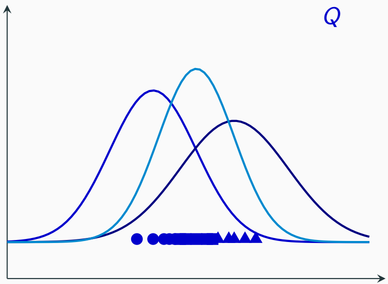

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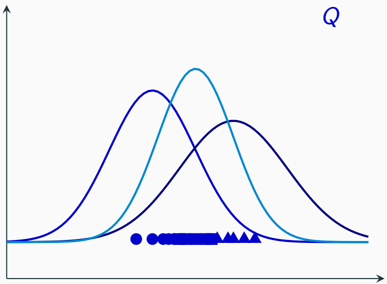

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Notation: $\mathbf{X}_{n,m} \sim Q$

Hierarchical Samples in the Literature

- Baxter (2000) — meta-learning, distribution over tasks
- Szabó et al. (2016) — distribution regression
- Haviv et al. (2024) — generative modelling over distributions
- Warren et al. (2025) — principal curves in Wasserstein space

Limitations of the Hierarchical Sampling Framework

- Each row of $\mathbf{X}_{n,m}$ has the law

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- **But for** m **large:**

$$Q^1 = Q^2 \iff \mathbf{X}_{n,m} \overset{d}{=} \mathbf{Y}_{n,m}.$$

Distance-based Two Sample Test

- **Problem.** $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$; test $H_0 : Q^1 = Q^2$ vs $H_1 : Q^1 \neq Q^2$.

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- **Strategy.** Given hierarchical samples $\mathbf{X}_{n,m} \sim Q^1$ and $\mathbf{Y}_{n,m} \sim Q^2$, reject H_0 if $d(\mathbf{X}_{n,m}, \mathbf{Y}_{n,m})$ is large.

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- **Ingredients.** A distance d; a threshold.

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Metrics on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$: Wasserstein over Wasserstein

- $(\mathbb{X}, d_{\mathbb{X}})$ is a Polish space with bounded diameter.
- **Step 1:** Using $d_{\mathbb{X}}$ on $\mathbb{X}$, define Wasserstein distance $\mathcal{W}$ on $\mathcal{P}(\mathbb{X})$: for $P^1, P^2 \in \mathcal{P}(\mathbb{X})$,

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- **Step 2:** Using $\mathcal{W}$ on $\mathcal{P}(\mathbb{X})$, for $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

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$\mathcal{W}_{\mathcal{W}}$ is called **Wasserstein over Wasserstein (WoW).**

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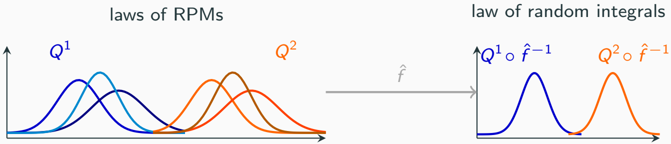

- **Hierarchical Integral Probability Metric** (Catalano and Lavenant 2024):

$$d_{\mathcal{F}}(Q^1, Q^2) = \sup_{f \in \mathcal{F}} \mathcal{W}\big(Q^1 \circ \hat{f}^{-1}, \, Q^2 \circ \hat{f}^{-1}\big).$$

- $\mathcal{F} = \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})$: $\quad d_{\mathcal{F}} = d_{\mathrm{Lip}}$.

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- $\mathbb{E}\big[\mathcal{W}_{\mathcal{W}}(\widehat{Q}_{n,m}, Q)\big] \to 0$ **slower than** any polynomial rate.
- $\mathbb{E}\big[d_{\mathrm{Lip}}(\widehat{Q}_{n,m}, Q)\big] \to 0$ at **polynomial rate**.

(Catalano and Lavenant 2024)

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$$\mathbf{X}_{n,m} \sim Q^1, \qquad \mathbf{Y}_{n,m} \sim Q^2.$$

- Build hierarchical empirical estimators:

$$\widehat{Q}^1_{n,m} = \frac{1}{n} \sum_{i=1}^{n} \delta_{\widehat{P}^1_{i,m}}, \qquad \widehat{Q}^2_{n,m} = \frac{1}{n} \sum_{i=1}^{n} \delta_{\widehat{P}^2_{i,m}}.$$

- **Reject** H_0 if

$$d\left(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}\right) \text{ is large}.$$

Theoretical Threshold via Rademacher Complexity

- **Idea:** Threshold $r(n, m, \theta)$ from a concentration inequality.

Theoretical Threshold via Rademacher Complexity

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Theorem
Let $\mathcal{G}$, $\mathcal{H}$ be uniformly bounded function classes generating IPMs on $\mathcal{P}(\mathbb{X})$ and $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ respectively, with $\mathcal{H} \subseteq \mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{I}_{\mathcal{G}})$. Then for all $\theta \in (0, 1)$, $n, m \geq 1$,

$$\mathbb{P}\Big(\big|\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - \mathcal{I}_{\mathcal{H}}(Q^1, Q^2)\big| \geq r(n, m, \theta)\Big) \leq \theta,$$

where

$$r(n, m, \theta) = O\left(\mathcal{R}_n(\mathcal{H}) + \mathcal{R}_m(\mathcal{G}) + \frac{1}{\sqrt{n}}\right).$$

Theoretical Threshold via Rademacher Complexity

$\mathbb{X} \subset \mathbb{R}^d$ bounded.

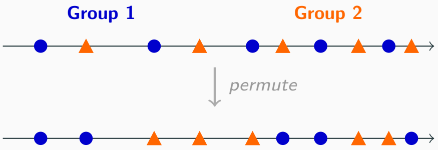
WoW:

$$\begin{cases} \mathcal{O}\left(\frac{\log\log n}{\log n} + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & d = 1 \\ \mathcal{O}\left(\frac{\log\log n}{\log n} + \frac{\log n}{\sqrt{n}} + \frac{\log m}{\sqrt{m}}\right), & d = 2 \\ \mathcal{O}\left(\frac{\log\log n}{\log n} + \frac{1}{n^{1/d}} + \frac{1}{m^{1/d}}\right), & d > 2. \end{cases}$$

HIPM:

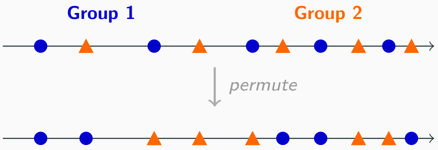
$$\begin{cases} \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & d = 1 \\ \mathcal{O}\left(\frac{\log n}{\sqrt{n}} + \frac{\log m}{\sqrt{m}}\right), & d = 2 \\ \mathcal{O}\left(\frac{1}{n^{1/d}} + \frac{1}{m^{1/d}}\right), & d > 2. \end{cases}$$

Practical Threshold via Permutation



Practical Threshold via Permutation

- Observe $d\left(\widehat{Q}^1_{n,m},\ \widehat{Q}^2_{n,m}\right)$. **Idea:** Estimate its $(1-\theta)$-quantile under H_0.



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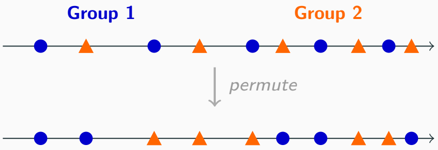
- $\widehat{P}^1_{1,m}, \ldots, \widehat{P}^1_{n,m},\ \widehat{P}^2_{1,m}, \ldots, \widehat{P}^2_{n,m} \overset{\text{i.i.d}}{\sim}$ under H_0



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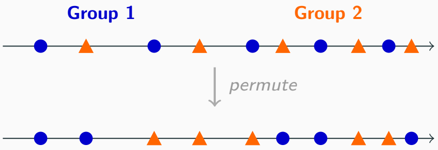
- For $s = 1, \ldots, S$: compute $d\left(\widehat{Q}^{1,(s)}_{n,m},\ \widehat{Q}^{2,(s)}_{n,m}\right)$



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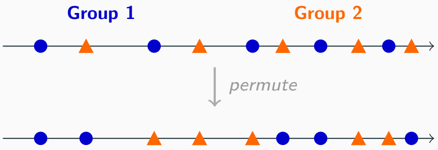
- For $s = 1, \ldots, S$: compute $d\left(\widehat{Q}^{1,(s)}_{n,m},\ \widehat{Q}^{2,(s)}_{n,m}\right)$
- **Threshold:** $(1-\theta)$-quantile of

$$d\left(\widehat{Q}^{1,(1)}_{n,m},\ \widehat{Q}^{2,(1)}_{n,m}\right), \ldots, d\left(\widehat{Q}^{1,(S)}_{n,m},\ \widehat{Q}^{2,(S)}_{n,m}\right)$$



Practical Threshold via Permutation

- **Test statistic:** $T_{n,m} = \sqrt{\frac{n}{2}}\, d_{\mathrm{Lip}}\left(\widehat{Q}^1_{n,m},\ \widehat{Q}^2_{n,m}\right)$
- **Permutation threshold:** for significance level $\theta \in (0,1)$,

$$r(n,m,\theta) = \inf\left\{ t \in \mathbb{R} : \mathbb{P}\left(\widetilde{T}_{n,m} > t \;\middle|\; \mathbf{X}_{n,m}, \mathbf{Y}_{n,m} \right) \leq \theta \right\}.$$

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- **Testing scheme:** Reject H_0 if $T_{n,m} > r(n,m,\theta)$.

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Theorem
Let $\mathbb{X} = [a,b]$ and $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Then for all $\theta \in (0,1)$

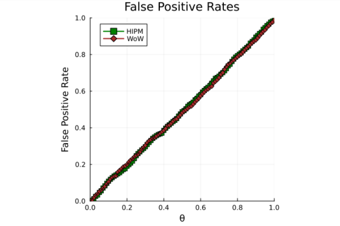
- *if $Q^1 = Q^2$:* $\lim_{n,m\to\infty} \mathbb{P}(T_{n,m} > r(n,m,\theta)) \leq \theta$.
- *if $Q^1 \neq Q^2$:* $\lim_{n,m\to\infty} \mathbb{P}(T_{n,m} > r(n,m,\theta)) = 1$.

WoW vs. HIPM: Type I error and Power

$n = m = n_{\mathrm{perm}} = 100.$

Type I error

$\mathrm{DP}(\alpha = 1, P_0 = \mathcal{N}(0,1))$



- Both methods **control Type I error**, including $\mathbb{X} = \mathbb{R}$ where theory does not guarantee it.

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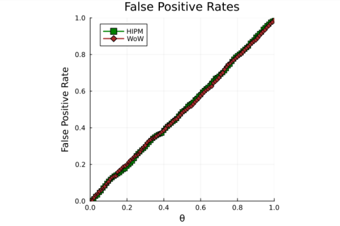
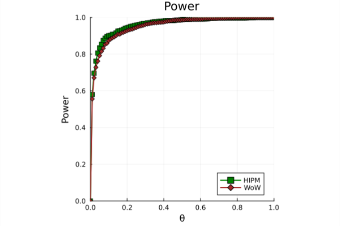


Power: A.

$\mathrm{DP}(\alpha = 1, P_0 = \mathrm{Beta}(a, b))$, varying P_0



- Both methods **control Type I error**, including $\mathbb{X} = \mathbb{R}$ where theory does not guarantee it.
- **A:** HIPM slightly outperforms WoW.

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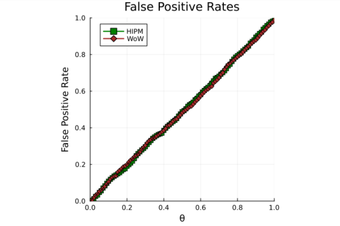

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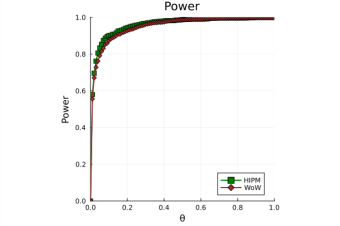
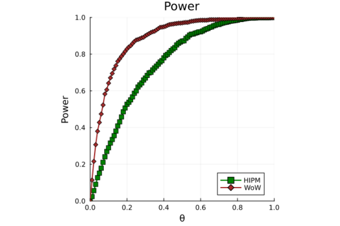


Power: B.

$\mathrm{DP}(\alpha = 1, P_0 = \mathcal{N}(0,1))$,
varying α



- Both methods **control Type I error**, including $\mathbb{X} = \mathbb{R}$ where theory does not guarantee it.
- **A:** HIPM slightly outperforms WoW.
- **B:** WoW outperforms HIPM.

Methods from the Literature

- **Fréchet test** (Dubey and Müller 2019)
- **Energy test** (Székely and Rizzo 2004)

Comparison with Existing Methods

Model: $\widetilde{P}^i|\widetilde{\mu} = \mathcal{N}(\widetilde{\mu}, 1)$, $\widetilde{\mu} \sim \mathcal{N}_T(\mu_{0i}, \sigma^2_{0i}, -10, 10)$. $n = 100$, $m = 200, \theta = 0.05$.

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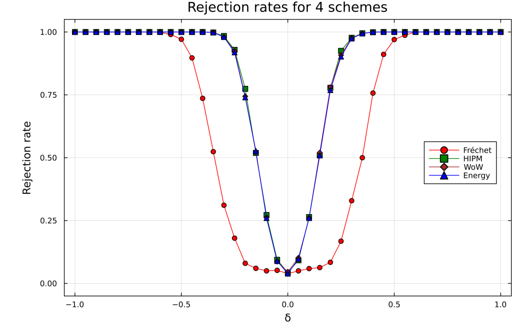
Mean shift: $\sigma_{0i} = 0.5, \mu_{01} = 0, \mu_{02} = \delta$



- **Mean shift:** Energy, WoW, HIPM outperform Fréchet.

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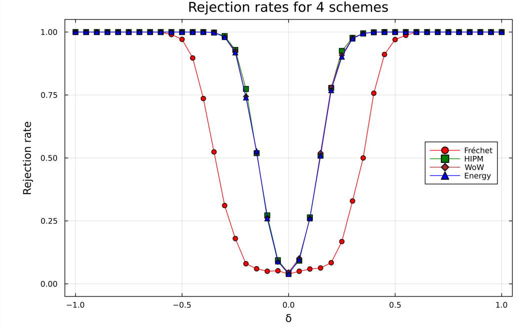
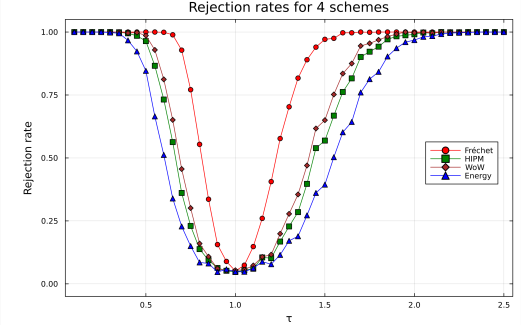


Variance shift: $\mu_{0i} = 0, \sigma_{01} = 0.2, \sigma_{02} = 0.2\tau$



- **Mean shift:** Energy, WoW, HIPM outperform Fréchet.
- **Variance shift:** Fréchet > WoW > HIPM > Energy.
- All methods control Type I error ($\delta = 0$, $\tau = 1$).

Comparison with Existing Methods

- $Q^1 \neq Q^2$ but **same** Fréchet mean and variance.

$\widetilde{P}|\widetilde{\mu} = \mathcal{N}(\widetilde{\mu}, 1)$ in both cases, with

$$Q^1: \ \widetilde{\mu} \sim \tfrac{1}{2}\delta_{-1} + \tfrac{1}{2}\delta_1,$$
$$Q^2: \ \widetilde{\mu} \sim \tfrac{1}{8}\delta_{-2} + \tfrac{3}{4}\delta_0 + \tfrac{1}{8}\delta_2.$$

Test family: $(Q^1,\, M_\lambda)$, $M_\lambda = \lambda Q^1 + (1-\lambda)Q^2$, $\lambda \in [0,1]$.

$\lambda = 1$: Type I error; $\lambda < 1$: power.

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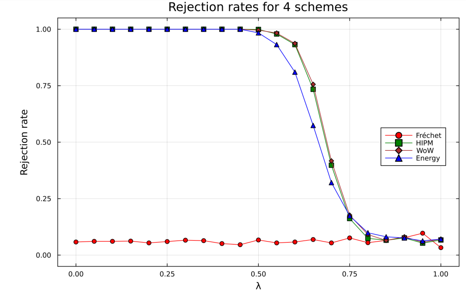


- Fréchet test cannot detect $Q^1 \neq M_\lambda$.
- HIPM, WoW, and Energy correctly detect the discrepancy.

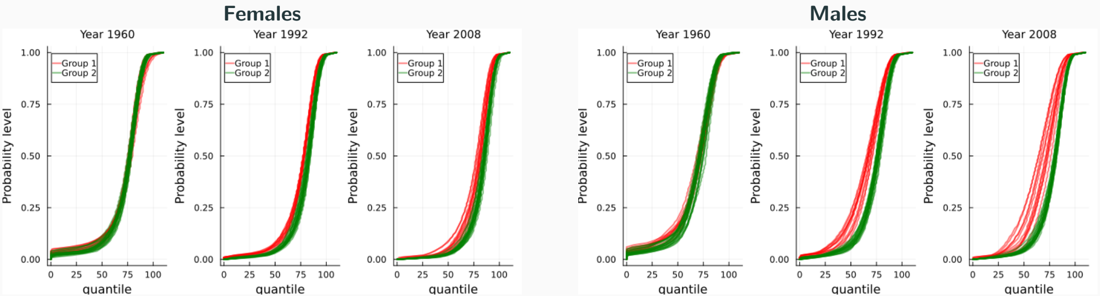
Application: Human Mortality Data

Data: Human Mortality Database, 1960–2010.

age-at-death distribution over ages $\{0, 1, \ldots, 110\}$ for each country and year.

- Group 1: Belarus, Bulgaria, Czechia, Estonia, Hungary, Latvia, Lithuania, Poland, Russia, Slovakia, Ukraine
- Group 2: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Iceland, Ireland, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Spain, Sweden, Switzerland, UK, USA.

Application: Human Mortality Data



Quantile plots of age-at-death distributions for years 1960, 1992, 2008.

- The years capture the Cold War era, post-Soviet collapse, and a more recent period.
- Differences are visually apparent in later periods.

Application: Human Mortality Data

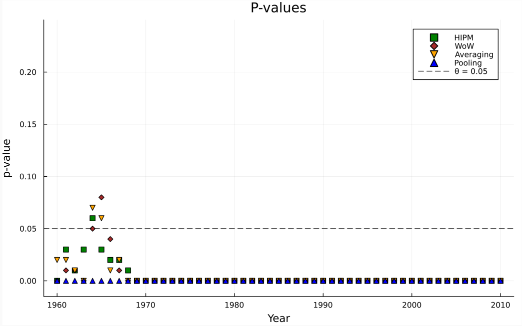
Methods:

- **HIPM, WoW**
- **Averaging**: $\overline{P}^k = \frac{1}{n_k}\sum_i \widehat{P}_i^k$; $\mathcal{W}_1(\overline{P}^1, \overline{P}^2)$.
- **Pooling**: aggregate all deaths per group; permute at observation level.

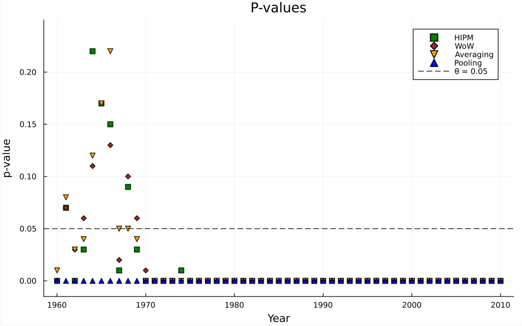
$\theta = 0.05$, $n_{\mathrm{perm}} = 100$

Application: Human Mortality Data

Males



Females



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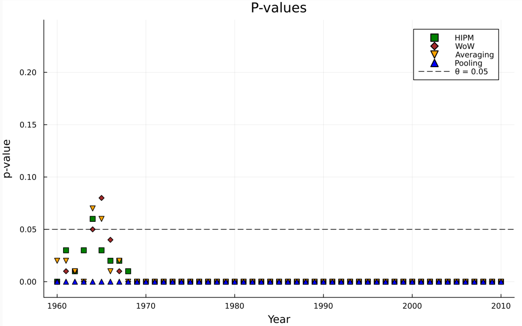

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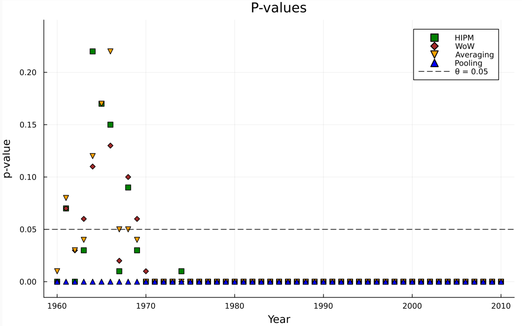

- HIPM, WoW, and Averaging show similar trends: significant differences after the 1970s.

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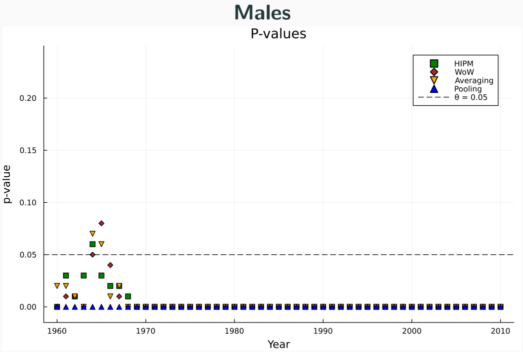

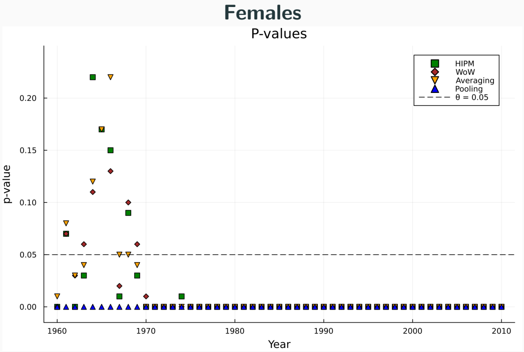

- HIPM, WoW, and Averaging show similar trends: significant differences after the 1970s.
- **Pooling** fails to control Type I error in hierarchical sampling setting.

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Thank you for your attention!